

116 Moo.2 Sridongyen, Chaiprakarn
Chiang Mai, Thailand 50320
Tel/Line ID: 0992419916
E-mail: ounjai.sa@gmail.com

**SARAWUT OUNJAI**

**Student of Development and Promotion of Science and Technology Talents Project (Royal Government of Thailand scholarship)**

## PROFESSIONAL SUMMARY

Data Science & ML Expert | PhD graduate in Life Sciences from Imperial College London with specialisation in machine learning, predictive analytics, and automated system development. Developed high-precision ML models achieving 99.98% accuracy in ASV authentication and processed over 170,000 samples. Proven expertise in developing AI algorithms for complex pattern recognition, big data management, and statistical forecasting across interdisciplinary domains. Strong research background with peer-reviewed publications demonstrating ability to translate complex data into actionable insights.

**Seeking opportunities in:** Data Science • AI/ML Engineering • Research & Development • Predictive Analytics • Biotechnology

## MOTIVATION

My passion for data science stems from discovering the transformative power of machine learning and predictive analytics in solving complex problems across diverse domains. Throughout my PhD, I found that developing automated systems and AI algorithms that extract meaningful insights was more rewarding than simply generating data. What excites me most is the universal applicability of data science—the statistical modeling, predictive analytics, and data management skills I've developed can create value in various industries, including financial institutions, government agencies, healthcare, and technology companies. I am motivated by the opportunity to translate complex data into actionable intelligence that drives evidence-based decision-making and creates a meaningful impact across industries.

## EDUCATION

**• 2018 – 2025: PhD Life Sciences**
Department of Life Sciences, Imperial College London, United Kingdom
Thesis: "Biodiversity of beetles: The phylogenetic composition of beetle communities across tropical forests of the world"

**• 2013 – 2015: M.Sc. Biology, GPA: 4.00**
Department of Biology, Faculty of Science, Chiang Mai University, Thailand
Thesis: "Identification of Medicinal Plants in Zingiberaceae by Barcode DNA High-Resolution Melting Analysis (Bar-HRM) Method"

**• 2009 – 2012: B.Sc. Biology, GPA: 3.60 (1st Class Honours)**
Department of Biology, Faculty of Science, Chiang Mai University, Thailand
Final Project: "The comparison and optimisation of secretion and export of hIGF-1 in Escherichia coli different strains"
**Awards:** Academic Excellence Awards

## TRAINING & PROFESSIONAL DEVELOPMENT

**2025: AI Basics | AI Governance & Ethics | AI for Everyone | Leadership in the Age of AI | Prompt Engineering for AI:** Digital Government Development Agency & Thailand Digital Government Academy

**2025: Introduction to AI:** Chiang Mai University

**2024: Elements of AI:** University of Helsinki (online)

**2015: Metabarcoding Training - Molecular Plant Identification**
Department of Organismal Biology, Systematic Biology, Uppsala University, Sweden

- Advanced training in DNA metabarcoding techniques for plant species identification
- High-throughput sequencing data analysis and bioinformatics workflows

**2012: Bar-HRM Technique Training**
Institute of Agrobiotechnology Centre for Research and Technology Hellas - CERTH, Greece

- Specialised training in Barcode DNA high-resolution melting analysis
- Applied molecular techniques for species authentication and identification

## TECHNICAL SKILLS & EXPERTISE

**Machine Learning & AI Development**

- **High-precision ASV authentication systems achieving 99.98% accuracy** for DNA barcoding & metabarcoding with over 170,000 ASV samples
  - Reduced manual analysis time from years to hours
- Image-based beetle identification algorithms using computer vision
- Taxonomic classification pipelines with automated pattern recognition
- Species-specific detection and identification systems

**Bioinformatics & Programming**

- **Programming Languages:** R programming, Python for data analysis and automation
- **Pipeline Development:** Automated DNA barcoding and metabarcoding processing systems
- **Phylogenetic Analysis:** Automated phylogenetic reconstruction tools and workflows
- **Molecular Tools:** Primer and probe design automation platforms
- **High-throughput Processing:** Sequencing data analysis and workflow optimisation

**Data Analytics & Management**

- **Statistical Analysis:** Advanced statistical modelling, phylogenetic reconstruction, biodiversity assessment
- **Predictive Modelling:** Statistical forecasting models for trend prediction and pattern analysis
- **Database Management:** Automated molecular biology data processing pipelines
- **Business Intelligence:** Automated reporting systems, data retrieval and management
- **Big Data:** High-throughput sequencing analysis, ecological data interpretation

**Molecular Biology & Laboratory Techniques**

- **DNA/RNA Techniques:** Extraction, purification, PCR, gel electrophoresis, cloning
- **Advanced Methods:** DNA barcoding, metabarcoding, metamitogenomics
- **Specialised Techniques:** Cytogenetics, Bar-HRM analysis
- **Quality Control:** Contamination filtering, pseudogene detection

**Software & Technical Tools**

- **Development Tools:** Visual Studio Code, Terminal, Git/GitHub
- **Bioinformatics Software:** MEGA, Geneious, Dendroscope, RAxML, Aliview
- **Database Systems:** Database design and management platforms
- **Geospatial Analysis:** QGIS for ecological and geographical data
- **Office Suite:** Microsoft Office for documentation and presentations

**Professional & Research Skills**

- **Project Management:** Cross-functional team collaboration, research project coordination
- **Communication:** Scientific writing, technical presentations to diverse stakeholders
- **International Experience:** Cross-cultural collaboration in research environments
- **Mentoring:** Student supervision and knowledge transfer
- **Adaptability:** Working effectively in interdisciplinary and international settings

## LINKS

- LinkedIn: https://www.linkedin.com/in/sarawut-ounjai-57701757
- Google Scholar: https://scholar.google.com/citations?hl=th&user=nlnxKH0AAAAJ
- GitHub: https://github.com/OunjaiS

## RESEARCH EXPERIENCES

**2025-present: Post-Doctoral Research - eDNA Monitoring of Invasive Fish Species:**
Chiang Mai University, Thailand

- Developed eDNA detection protocols for silver carp (*Hypophthalmichthys molitrix*) monitoring
- Established common carp detection systems in Thai river systems using environmental DNA
- Contributed to biodiversity monitoring and invasive species management strategies

**2018 – 2025: Doctoral Research - Biodiversity Assessment of Tropical Forest Beetles:**
Natural History Museum & Imperial College London, UK

- Developed comprehensive phylogenetic analysis of beetle communities across global tropical forests
- Implemented machine learning algorithms for automated beetle identification from morphological images
- Created automated bioinformatics pipelines for DNA barcoding and metabarcoding analysis (ASV authentication)
- Developed automated molecular biology processing pipeline (data management, retrieval, sequencing BLAST, alignment)
- Designed and optimised automated primer sets for species-specific detection

**2014-2017: Medicinal Plant Authentication Research:**
Chiang Mai University, Thailand

- Specialised in DNA barcoding coupled with high-resolution melting analysis (Bar-HRM)
- Conducted authentication studies of Thai herbal products and Zingiberaceae species
- Developed rapid discrimination methods for closely related plant species

**2012-2013: Recombinant Protein Production Research:**
Chiang Mai University, Thailand

- Investigated recombinant insulin-like growth factor 1 (IGF-1) production in E. coli strains
- Optimised protein secretion and export systems in bacterial expression systems

## PUBLICATIONS

**Published Articles**

- **Ounjai, S.**, Osathanunkul, R., Madesis, P. & Osathanunkul, M. (2017). Multi chloroplast genes for species identification in Bar-HRM analysis of taxonomical complex medicinal plants group (Zingiberaceae). Chiang Mai Journal of Science. CMJS.06.01.16-6559.
- de Boer, H.J., Ghorbani, A., Manzanilla, V., Raclariu, A.C., Kreziou, A., **Ounjai, S.**, Osathanunkul, M., Gravendeel, B. (2017). DNA metabarcoding of orchid-derived products reveals widespread illegal orchid trade. Proc Biol Sci 284.
- Osathanunkul, M., Madesis, P., **Ounjai, S.**, Pumiputavon, K., Somboonchai, R., Lithanatudom, P., & Suwannapoom, C. (2016). Identification of Uvaria sp by barcoding coupled with high-resolution melting analysis (Bar-HRM). Genetics and Molecular Research, 15(1).
- Osathanunkul, M., Madesis, P., **Ounjai, S.**, Suwannapoom, C., & Jampeetong, A. (2015). Rapid discrimination between four seagrass species using hybrid analysis. Genetics and Molecular Research, 14(2), 3957-3963.
- Osathanunkul, M., Pintakum, D., **Ounjai, S.**, Madesis, P., & Suwannapoom, C. (2015). Authenticating Thai herbal products, Boraphet: Tinospora crispa (Menispermeae) by DNA barcoding coupled with high resolution melting analysis. Genome, 58(5), 263-263.
- Osathanunkul, M., Suwannapoom, C., **Ounjai, S.**, Rora, J. A., Madesis, P., & de Boer, H. (2015). Refining DNA Barcoding Coupled High Resolution Melting for Discrimination of 12 Closely Related Croton Species. PLoS One, 10(9).
- Buddhachat, K., **Ounjai, S.**, Chokepaichitkool, T., Osathanunkul, M., (2012). The Comparison of Secretion and Export of hIGF-1 in Escherichia Coli Strain BL21 and DH5α. Journal of biomolecular techniques, 23(Suppl):S57-S57.

**Manuscripts Submitted/Under Review**

- Tracing the rise of silver carp (*Hypophthalmichthys molitrix*) in Thailand with eDNA. *NeoBiota* (submitted, 2025).
- Catching common carp with eDNA in Thailand's rivers. *iScience* (submitted, 2025).

**Manuscripts in Preparation**

- ASV Selection and Taxonomic Assignment in high-throughput DNA barcoding: Filtering of contaminants and pseudogenes. *Molecular Ecology Resources* (in preparation, 2025).
- Site-Based Assessment of Beetle Diversity in Central Thailand. *Ecological Entomology* (in preparation, 2025).
- Urbanisation Reshapes River Fish Communities Tracked Through eDNA. *Journal of Environmental Management* (in preparation, 2025).

## RESEARCH IMPACT & METRICS

- Total Citations: 164 (Google Scholar)
- h-index: 4
- i10-index: 4

## REFERENCES


• **Professor Dr. Alfried Vogler**
Department of Life Sciences, Faculty of Natural Sciences
Imperial College London, London SW7 2AZ, UK

• **Associate Professor Dr. Maslin Osathanunkul**
Department of Biology, Faculty of Science
Chiang Mai University, Chiang Mai 50200, Thailand

• **Associate Professor Dr. Kittisak Buddhachat**
Department of Biology, Faculty of Science
Naresuan University, Phitsanulok 65000, Thailand